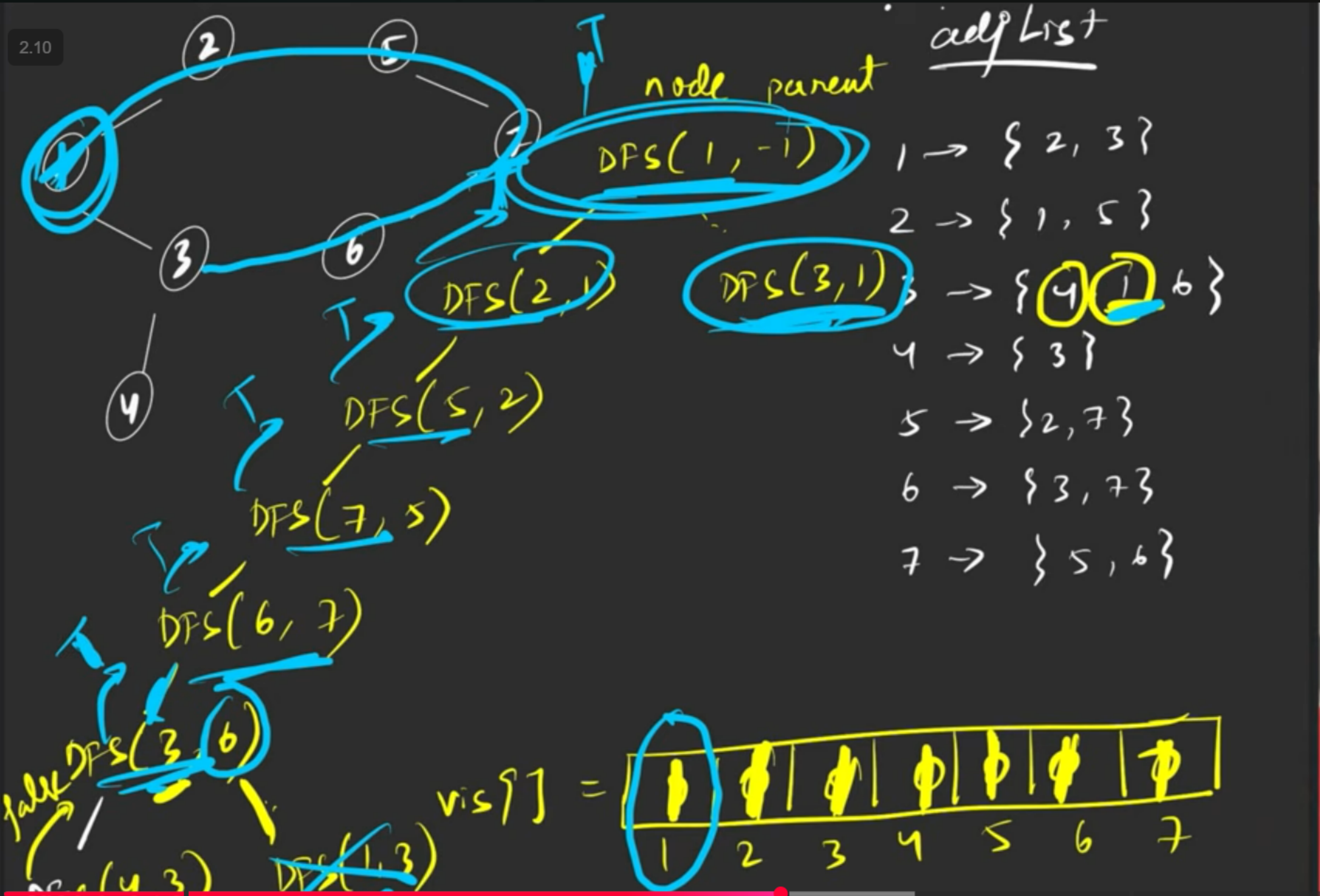
G-12. Detect a Cycle in an Undirected Graph using DFS | C++ | Java



8:22 / 19:09

DFS

```
dfs(node, parent)
{
    vis[node] = 1

    for (auto it : adj[node])
    {
        if (vis[it] == 0)
        {
            if (dfs(it, node) == true)
                return True;
        }

        else if (it != parent)
            return True;

    }

ret. false;
```

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```java
class Solution {
    private boolean dfs(int node, int parent,
                        int vis[],
                        ArrayList<ArrayList<Integer>> adj) {
        vis[node] = 1;
        for(int adjacentNode: adj.get(node)) {
            if(vis[adjacentNode]==0) {
                if(dfs(adjacentNode, node, vis, adj) == true)
                    return true;
            }
            else if(adjacentNode != parent) return true;
        }
        return false;
    }
    // Function to detect cycle in an undirected graph.
    public boolean isCycle(int V,
                           ArrayList<ArrayList<Integer>> adj) {
       int vis[] = new int[V];
       for(int i = 0;i<V;i++) {
           if(vis[i] == 0) {
               if(dfs(i, -1, vis, adj) == true) return true;
           }
       }
       return false;
    }
}
```

cycle in the graph.

Task:

C++ (g++ 5.4)

```cpp
// } Driver Code Ends
class Solution {
  private:
    bool dfs(int node, int parent, vector<int> vis[], vector<int> adj[]) {
        vis[node] = 1;
        for(auto adjacentNode: adj[node]) {
            if(!vis[adjacentNode]) {
                if(dfs(adjacentNode, node, vis, adj) == true)
                    return true;
            }
            else if(adjacentNode != parent) return true;
        }
        return false;
    }
  public:
    // Function to detect cycle in an undirected graph.
    bool isCycle(int V, vector<int> adj[]) {
       int vis[V] = {0};
       return dfs(1, -1, vis, adj);
    }
};
```


```
for (i = 1 → 9)
{ if (!vis[i])
      dfs(i) == T
        return T;
}
```

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```java
class Solution {
    private boolean dfs(int node, int parent,
                        int vis[],
                        ArrayList<ArrayList<Integer>> adj) {
        vis[node] = 1;
        for(int adjacentNode: adj.get(node)) {
            if(vis[adjacentNode]==0) {
                if(dfs(adjacentNode, node, vis, adj) == true)
                    return true;
            }
            else if(adjacentNode != parent) return true;
        }
        return false;
    }
    // Function to detect cycle in an undirected graph.
    public boolean isCycle(int V,
                        ArrayList<ArrayList<Integer>> adj) {
       int vis[] = new int[V];
       for(int i = 0;i<V;i++) {
           if(vis[i] == 0) {
               if(dfs(i, -1, vis, adj) == true) return true;
           }
       }
       return false;
    }
}
```

C++ (g++ 5.4)

```cpp
// } Driver Code Ends
class Solution {
  private:
    bool dfs(int node, int parent, vector<int> vis[], vector<int> adj[])
        vis[node] = 1;
        for(auto adjacentNode: adj[node]) {
            if(!vis[adjacentNode]) {
                if(dfs(adjacentNode, node, vis, adj) == true)
                    return true;
            }
            else if(adjacentNode != parent) return true;
        }
        return false;
    }
  public:
    // Function to detect cycle in an undirected graph.
    bool isCycle(int V, vector<int> adj[]) {
       int vis[V] = {0};
       for(int i = 0;i<V;i++) {
           if(!vis[i]) {
               if(dfs(i, -1, vis, adj) == true) return true;
           }
       }
       return false;
    }
};
```

16:58 / 19:09 | Code

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dfs( )
return T;
}

SC → $O(N) + O(N) \approx O(N)$

TC → $O(N + 2E) + O(N)$

18:34 / 19:09 Code